MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 14: Linear Hyperbolic Systems and the Riemann Problem

Instructor: Biswarup Biswas

1 Introduction: From One Equation to Several

Lectures 10 to 13 built a complete toolbox for a *scalar* conservation law $u_t + f(u)_x = 0$: finite volumes, Godunov's method, MUSCL, TVD limiters, ENO and WENO. Every one of these methods rests on a single idea. Information travels along characteristics with speed $f'(u)$, and a good scheme takes its data from the *upwind* side.

The equations of fluid flow are not scalar. The one-dimensional Euler equations of Lecture 2 are three coupled equations for mass, momentum and energy, and along them travel three kinds of waves: two sound waves and one contact (entropy) wave. Two new questions arise:

1. What does "upwind" mean when there are several waves, some moving left and some moving right?
2. What is the solution of the Riemann problem, which Godunov's method needs at every interface?

This lecture answers both questions for **linear** systems with constant coefficients. The answer is clean: an eigenvector decomposition splits the system into independent scalar advection equations, one per wave, and everything we know about the scalar case applies to each of them. Linear systems are also the model for the nonlinear case: Roe's approximate Riemann solver (Module 6) linearizes the Euler equations at every interface and then does exactly what we do in this lecture.

2 Systems of Conservation Laws

2.1 Conservative and Quasi-Linear Forms

A system of m conservation laws in one space dimension reads

$$\mathbf{q}_t + \mathbf{f}(\mathbf{q})_x = \mathbf{0}, \qquad \mathbf{q}(x,t) = \begin{pmatrix} q_1 \\ \vdots \\ q_m \end{pmatrix} \in \mathbb{R}^m, \qquad \mathbf{f}(\mathbf{q}) = \begin{pmatrix} f_1(\mathbf{q}) \\ \vdots \\ f_m(\mathbf{q}) \end{pmatrix}. \tag{1}$$

Each component q_i is a conserved quantity and f_i is its flux. For smooth solutions we can differentiate the flux by the chain rule. Each f_i depends on x through all of $q_1, \ldots, q_m$, so

$$\frac{\partial}{\partial x} f_i\big(\mathbf{q}(x,t)\big) = \frac{\partial f_i}{\partial q_1}\frac{\partial q_1}{\partial x} + \frac{\partial f_i}{\partial q_2}\frac{\partial q_2}{\partial x} + \cdots + \frac{\partial f_i}{\partial q_m}\frac{\partial q_m}{\partial x} = \sum_{j=1}^{m} \frac{\partial f_i}{\partial q_j}\,(q_j)_x.$$

Collecting the m rows into matrix form gives $\mathbf{f}(\mathbf{q})_x = A(\mathbf{q})\,\mathbf{q}_x$, and (1) becomes the **quasi-linear form**

$$\mathbf{q}_t + A(\mathbf{q})\,\mathbf{q}_x = \mathbf{0}, \qquad A(\mathbf{q}) = \mathbf{f}'(\mathbf{q}) = \begin{pmatrix} \frac{\partial f_1}{\partial q_1} & \cdots & \frac{\partial f_1}{\partial q_m} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial q_1} & \cdots & \frac{\partial f_m}{\partial q_m} \end{pmatrix}. \tag{2}$$

$A(\mathbf{q})$ is the $m \times m$ **flux Jacobian**. It plays the role of $f'(u)$ in the scalar case. When $\mathbf{f}(\mathbf{q}) = A\mathbf{q}$ with a constant matrix A, the Jacobian is A itself and the system is **linear**: $\mathbf{q}_t + A\mathbf{q}_x = \mathbf{0}$.

Example: the Euler equations. In one dimension, with $\mathbf{q} = (\rho, \rho u, E)^T$,

$$\mathbf{f}(\mathbf{q}) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E+p) \end{pmatrix}, \qquad p = (\gamma - 1)\left(E - \tfrac{1}{2}\rho u^2\right). \tag{3}$$

The Jacobian $A(\mathbf{q})$ is a 3×3 matrix whose entries depend on $\mathbf{q}$. Its eigenvalues are $u - c$, u and $u + c$, where $c = \sqrt{\gamma p/\rho}$ is the speed of sound (Exercise E6). We shall work them out fully in Module 6. In this lecture we study a smaller system with the same sound waves.

2.2 A Linear Model: Acoustics

We derive the equations of sound by linearizing the isentropic Euler equations about a uniform background state.

Step 1: Isentropic Euler in conservative form. If the flow is isentropic, the pressure is a function of density only, $p = P(\rho)$, and the energy equation is not needed. Mass and momentum conservation read

$$\rho_t + (\rho u)_x = 0, \qquad (\rho u)_t + (\rho u^2 + p)_x = 0. \tag{4}$$

Step 2: Rewrite in the variables (p, u). *Mass equation.* Expand the product: $(\rho u)_x = u\rho_x + \rho u_x$, so

$$\rho_t + u\rho_x + \rho u_x = 0. \tag{5}$$

Momentum equation. Expand both products:

$$(\rho u)_t = \rho u_t + u\rho_t, \qquad (\rho u^2)_x = (\rho u \cdot u)_x = u(\rho u)_x + \rho u\, u_x.$$

Substituting,

$$\rho u_t + u \underbrace{\left[\rho_t + (\rho u)_x\right]}_{=0 \text{ by mass}} + \rho u\, u_x + p_x = 0 \quad \Longrightarrow \quad u_t + u\, u_x + \frac{1}{\rho} p_x = 0.$$

Pressure equation. Define $c^2 = P'(\rho)$ (we shall see that c is the sound speed). By the chain rule, $p_t = P'(\rho)\rho_t = c^2\rho_t$ and $p_x = c^2\rho_x$. Multiplying (5) by c^2:

$$c^2\rho_t + u\, c^2\rho_x + \rho c^2 u_x = 0 \quad \Longrightarrow \quad p_t + u\, p_x + \rho c^2\, u_x = 0.$$

So the isentropic Euler equations in the variables (p, u) are

$$p_t + u\, p_x + \rho c^2\, u_x = 0, \qquad u_t + u\, u_x + \frac{1}{\rho} p_x = 0. \tag{6}$$

Step 3: Linearize. Let (ρ_0, u_0, p_0) be a constant background state and write

$$\rho = \rho_0 + \rho', \qquad u = u_0 + u', \qquad p = p_0 + p',$$

where the primed perturbations are small. Since ρ_0, u_0, p_0 are constants, $p_t = p'_t$, $p_x = p'_x$, $u_t = u'_t$, $u_x = u'_x$. Term by term in (6):

$$\begin{aligned}
u\, p_x &= (u_0 + u')\, p'_x = u_0\, p'_x + \underbrace{u' p'_x}_{\text{small}\times\text{small}} \approx u_0\, p'_x, \\
\rho c^2\, u_x &= \left(\rho_0 c_0^2 + \text{small}\right) u'_x \approx \rho_0 c_0^2\, u'_x, \qquad c_0^2 = P'(\rho_0), \\
u\, u_x &= (u_0 + u')\, u'_x \approx u_0\, u'_x, \qquad \frac{1}{\rho} p_x = \frac{p'_x}{\rho_0 + \rho'} \approx \frac{p'_x}{\rho_0}.
\end{aligned}$$

In each line we dropped products of two small quantities. Dropping the primes, we obtain the equations of **linear acoustics**.

Linear Acoustics

$$\begin{aligned} p_t + u_0\, p_x + K_0\, u_x &= 0, \\ u_t + \tfrac{1}{\rho_0}\, p_x + u_0\, u_x &= 0, \end{aligned} \qquad \Longleftrightarrow \qquad \begin{pmatrix} p \\ u \end{pmatrix}_t + \underbrace{\begin{pmatrix} u_0 & K_0 \\ 1/\rho_0 & u_0 \end{pmatrix}}_{A} \begin{pmatrix} p \\ u \end{pmatrix}_x = \mathbf{0}, \tag{7}$$

with

$$c_0 = \sqrt{P'(\rho_0)} \text{ (sound speed)}, \quad K_0 = \rho_0 c_0^2 \text{ (bulk modulus)}, \quad Z_0 = \rho_0 c_0 \text{ (impedance)}.$$

u_0 is the background flow velocity. p and u are now the *perturbations*, so p can be negative.

Two identities between these constants will be used again and again:

$$K_0 = \rho_0 c_0^2 = (\rho_0 c_0)\, c_0 = Z_0\, c_0, \qquad \frac{1}{\rho_0} = \frac{c_0}{\rho_0 c_0} = \frac{c_0}{Z_0}. \tag{8}$$

Acoustics will be our running example throughout this lecture.

3 Hyperbolicity

3.1 Definition

In Lecture 3 we classified scalar second-order PDEs. For first-order systems the classification is through the eigenvalues of A.

Definition: Hyperbolic System

The linear system $\mathbf{q}_t + A\mathbf{q}_x = 0$ is **hyperbolic** if A is diagonalizable with real eigenvalues, that is,

$$A = R\,\Lambda\,R^{-1}, \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_m), \quad \lambda_p \in \mathbb{R}, \tag{9}$$

where the columns $\mathbf{r}_1, \ldots, \mathbf{r}_m$ of R are linearly independent eigenvectors, $A\mathbf{r}_p = \lambda_p \mathbf{r}_p$. It is **strictly hyperbolic** if, in addition, the eigenvalues are distinct.
The nonlinear system (1) is hyperbolic if $A(\mathbf{q})$ has this property for every physically admissible state $\mathbf{q}$.

We order the eigenvalues as $\lambda_1 \le \lambda_2 \le \cdots \le \lambda_m$. The eigenvalue λ_p is the speed of the p-th **wave family**, and the eigenvector $\mathbf{r}_p$ tells us which combination of the variables jumps across a p-wave. (Distinct eigenvalues always give independent eigenvectors, so a strictly hyperbolic system is automatically hyperbolic.)

3.2 Eigenstructure of Acoustics

Eigenvalues. With A from (7),

$$\det(A - \lambda I) = \det \begin{pmatrix} u_0 - \lambda & K_0 \\ 1/\rho_0 & u_0 - \lambda \end{pmatrix} = (u_0 - \lambda)^2 - \frac{K_0}{\rho_0} = (u_0 - \lambda)^2 - c_0^2,$$

using $K_0/\rho_0 = c_0^2$. Setting this to zero, $u_0 - \lambda = \pm c_0$, so

$$\lambda_1 = u_0 - c_0, \qquad \lambda_2 = u_0 + c_0. \tag{10}$$

These are real and distinct (since $c_0 > 0$): acoustics is **strictly hyperbolic**. Sound travels at speed c_0 relative to the fluid, in both directions, and is carried along by the background flow u_0.

Eigenvector for $\lambda_1 = u_0 - c_0$. We solve $(A - \lambda_1 I)\mathbf{r} = \mathbf{0}$ for $\mathbf{r} = (r^{(1)}, r^{(2)})^T$. Since $u_0 - \lambda_1 = c_0$,

$$\begin{pmatrix} c_0 & K_0 \\ 1/\rho_0 & c_0 \end{pmatrix} \begin{pmatrix} r^{(1)} \\ r^{(2)} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The first row gives $c_0 r^{(1)} + K_0 r^{(2)} = 0$, i.e. $r^{(1)} = -\dfrac{K_0}{c_0} r^{(2)} = -Z_0\, r^{(2)}$ by (8). Eigenvectors are only defined up to a factor, so we choose $r^{(2)} = 1$: $\mathbf{r}_1 = (-Z_0, 1)^T$. The second row must then hold automatically. Check: $\frac{1}{\rho_0}(-Z_0) + c_0 \cdot 1 = -\frac{\rho_0 c_0}{\rho_0} + c_0 = 0$. ✓

Eigenvector for $\lambda_2 = u_0 + c_0$. Now $u_0 - \lambda_2 = -c_0$, and the first row of $(A - \lambda_2 I)\mathbf{r} = \mathbf{0}$ is $-c_0 r^{(1)} + K_0 r^{(2)} = 0$, so $r^{(1)} = Z_0 r^{(2)}$ and $\mathbf{r}_2 = (Z_0, 1)^T$.

Direct check. It is good practice to verify $A\mathbf{r}_1 = \lambda_1 \mathbf{r}_1$ by multiplying out:

$$A\mathbf{r}_1 = \begin{pmatrix} u_0 & K_0 \\ 1/\rho_0 & u_0 \end{pmatrix} \begin{pmatrix} -Z_0 \\ 1 \end{pmatrix} = \begin{pmatrix} -u_0 Z_0 + K_0 \\ -Z_0/\rho_0 + u_0 \end{pmatrix} = \begin{pmatrix} -u_0 Z_0 + Z_0 c_0 \\ -c_0 + u_0 \end{pmatrix} = (u_0 - c_0) \begin{pmatrix} -Z_0 \\ 1 \end{pmatrix}. \checkmark$$

The matrices R and R^{-1}. Putting the eigenvectors in the columns,

$$R = [\mathbf{r}_1 | \mathbf{r}_2] = \begin{pmatrix} -Z_0 & Z_0 \\ 1 & 1 \end{pmatrix}, \qquad \det R = (-Z_0)(1) - (Z_0)(1) = -2Z_0.$$

By the 2×2 inverse formula,

$$R^{-1} = \frac{1}{-2Z_0} \begin{pmatrix} 1 & -Z_0 \\ -1 & -Z_0 \end{pmatrix} = \frac{1}{2Z_0} \begin{pmatrix} -1 & Z_0 \\ 1 & Z_0 \end{pmatrix}. \tag{11}$$

Check: $RR^{-1} = \dfrac{1}{2Z_0} \begin{pmatrix} (-Z_0)(-1) + Z_0(1) & (-Z_0)Z_0 + Z_0 Z_0 \\ (1)(-1) + (1)(1) & (1)Z_0 + (1)Z_0 \end{pmatrix} = \dfrac{1}{2Z_0} \begin{pmatrix} 2Z_0 & 0 \\ 0 & 2Z_0 \end{pmatrix} = I.$ ✓

Eigenstructure of Linear Acoustics

$$\lambda_1 = u_0 - c_0, \quad \lambda_2 = u_0 + c_0, \qquad \mathbf{r}_1 = \begin{pmatrix} -Z_0 \\ 1 \end{pmatrix}, \quad \mathbf{r}_2 = \begin{pmatrix} Z_0 \\ 1 \end{pmatrix}, \qquad R^{-1} = \frac{1}{2Z_0} \begin{pmatrix} -1 & Z_0 \\ 1 & Z_0 \end{pmatrix}. \tag{12}$$

The eigenvectors do not depend on u_0; only the speeds do.

4 Characteristic Decomposition

4.1 Decoupling the System

Let the system be hyperbolic with constant $A = R\Lambda R^{-1}$. Define the **characteristic variables**

$$\mathbf{w} = R^{-1}\mathbf{q}, \qquad \text{equivalently} \qquad \mathbf{q} = R\,\mathbf{w} = w_1\mathbf{r}_1 + w_2\mathbf{r}_2 + \cdots + w_m\mathbf{r}_m. \tag{13}$$

So w_p is the coordinate of $\mathbf{q}$ along the eigenvector $\mathbf{r}_p$. Since R is constant, $\mathbf{q}_t = R\mathbf{w}_t$ and $\mathbf{q}_x = R\mathbf{w}_x$. Substituting into $\mathbf{q}_t + A\mathbf{q}_x = 0$:

$$R\,\mathbf{w}_t + AR\,\mathbf{w}_x = \mathbf{0}.$$

Multiply from the left by R^{-1} and use $R^{-1}AR = \Lambda$:

$$\mathbf{w}_t + \Lambda\,\mathbf{w}_x = \mathbf{0} \qquad \Longleftrightarrow \qquad (w_p)_t + \lambda_p\,(w_p)_x = 0, \quad p = 1, \ldots, m. \tag{14}$$

Because Λ is diagonal, row p involves only w_p. The coupled system has become m **independent scalar advection equations**, each of which we solved in Lecture 4: $w_p(x,t) = w_p(x - \lambda_p t, 0)$. Transforming back with $\mathbf{q} = R\mathbf{w}$:

Solution of a Linear Hyperbolic System

$$\mathbf{q}(x,t) = \sum_{p=1}^{m} w_p(x - \lambda_p t,\, 0)\, \mathbf{r}_p, \qquad \mathbf{w}(x,0) = R^{-1}\mathbf{q}(x,0). \tag{15}$$

Recipe: (1) compute $\mathbf{w}(x,0) = R^{-1}\mathbf{q}(x,0)$; (2) shift each w_p by $\lambda_p t$; (3) recombine $\mathbf{q} = \sum_p w_p \mathbf{r}_p$.

Domain of dependence. By (15), $\mathbf{q}(x,t)$ depends on the initial data only at the m points $x - \lambda_1 t, \ldots, x - \lambda_m t$. All these points lie in $[x - \lambda_m t,\ x - \lambda_1 t]$. Information propagates at finite speed, at most $\max_p |\lambda_p|$. This is why a CFL condition will appear for explicit methods (Section 6).

4.2 Acoustics: Left- and Right-Going Sound

The characteristic variables. From (12),

$$\mathbf{w} = R^{-1}\mathbf{q} = \frac{1}{2Z_0}\begin{pmatrix} -1 & Z_0 \\ 1 & Z_0 \end{pmatrix}\begin{pmatrix} p \\ u \end{pmatrix} = \frac{1}{2Z_0}\begin{pmatrix} -p + Z_0 u \\ p + Z_0 u \end{pmatrix}.$$

So, up to the constant factor $\pm\frac{1}{2Z_0}$, the characteristic variables are $p - Z_0 u$ (speed $u_0 - c_0$) and $p + Z_0 u$ (speed $u_0 + c_0$).

Direct verification. Let us check this without matrices, straight from (7). Using $p_t = -u_0 p_x - K_0 u_x$ and $u_t = -\frac{1}{\rho_0} p_x - u_0 u_x$,

$$\begin{aligned}
(p - Z_0 u)_t &= p_t - Z_0 u_t = -u_0 p_x - K_0 u_x + \frac{Z_0}{\rho_0} p_x + Z_0 u_0 u_x \\
&= -u_0\,(p - Z_0 u)_x + \frac{Z_0}{\rho_0} p_x - K_0 u_x = -u_0\,(p - Z_0 u)_x + c_0\, p_x - Z_0 c_0\, u_x \\
&= -u_0\,(p - Z_0 u)_x + c_0\,(p - Z_0 u)_x = -(u_0 - c_0)\,(p - Z_0 u)_x,
\end{aligned}$$

where we used $Z_0/\rho_0 = c_0$ and $K_0 = Z_0 c_0$. Hence $(p - Z_0 u)_t + (u_0 - c_0)(p - Z_0 u)_x = 0$. In the same way (Exercise E2) $(p + Z_0 u)_t + (u_0 + c_0)(p + Z_0 u)_x = 0$.

So $p - Z_0 u$ is constant along $dx/dt = u_0 - c_0$ and $p + Z_0 u$ is constant along $dx/dt = u_0 + c_0$. These are the acoustic **Riemann invariants**. Physically, a purely right-going wave has $w_1 = 0$, i.e. $p = Z_0 u$: pressure and velocity have the same sign, so a compression ($p > 0$) pushes the fluid forward ($u > 0$) in the direction the wave travels. A purely left-going wave has $p = -Z_0 u$.

4.3 Worked Example: A Pressure Bump Splits in Two

Take $u_0 = 0$ and initial data $p(x,0) = \phi(x)$, $u(x,0) = 0$, where ϕ is a bump.

Step 1. Characteristic variables at $t = 0$:

$$w_1(x,0) = \frac{-\phi(x) + Z_0 \cdot 0}{2Z_0} = -\frac{\phi(x)}{2Z_0}, \qquad w_2(x,0) = \frac{\phi(x) + 0}{2Z_0} = \frac{\phi(x)}{2Z_0}.$$

Step 2. Shift: $\lambda_1 = -c_0$ and $\lambda_2 = c_0$, so

$$w_1(x,t) = w_1(x + c_0 t, 0) = -\frac{\phi(x + c_0 t)}{2Z_0}, \qquad w_2(x,t) = w_2(x - c_0 t, 0) = \frac{\phi(x - c_0 t)}{2Z_0}.$$

Step 3. Recombine $\mathbf{q} = w_1 \mathbf{r}_1 + w_2 \mathbf{r}_2$, i.e. $p = -Z_0 w_1 + Z_0 w_2$ and $u = w_1 + w_2$:

$$p(x,t) = \tfrac{1}{2}\phi(x + c_0 t) + \tfrac{1}{2}\phi(x - c_0 t), \qquad u(x,t) = \frac{1}{2Z_0}\big[\phi(x - c_0 t) - \phi(x + c_0 t)\big].$$

The bump splits into two half-size copies moving apart at the speed of sound. The right-going half has $u = +p/Z_0$ and the left-going half has $u = -p/Z_0$, as predicted above.

Insight: Connection with the Wave Equation

This is d'Alembert's solution of the wave equation. Indeed, for $u_0 = 0$ differentiate the first equation of (7) in t and the second in x:

$$p_{tt} = -K_0\, u_{xt}, \qquad u_{tx} = -\tfrac{1}{\rho_0}\, p_{xx} \qquad \Longrightarrow \qquad p_{tt} = \frac{K_0}{\rho_0}\, p_{xx} = c_0^2\, p_{xx}.$$

A first-order hyperbolic system of two equations is a second-order wave equation in disguise, and c_0 is the wave speed.

5 The Riemann Problem for Linear Systems

5.1 Derivation of the Solution

The Riemann problem (Lecture 10) is the system with piecewise constant data,

$$\mathbf{q}(x,0) = \begin{cases} \mathbf{q}_L, & x < 0, \\ \mathbf{q}_R, & x > 0. \end{cases} \tag{16}$$

We apply the three-step recipe.

Step 1. The characteristic variables are also piecewise constant: $\mathbf{w}(x,0) = \mathbf{w}_L = R^{-1}\mathbf{q}_L$ for $x < 0$ and $\mathbf{w}_R = R^{-1}\mathbf{q}_R$ for $x > 0$. Define the jump in w_p,

$$\alpha_p = w_{p,R} - w_{p,L}, \qquad \text{i.e.} \qquad \boldsymbol{\alpha} = \mathbf{w}_R - \mathbf{w}_L = R^{-1}(\mathbf{q}_R - \mathbf{q}_L). \tag{17}$$

Multiplying by R, this is the same as expanding the jump in $\mathbf{q}$ along the eigenvectors:

$$\Delta\mathbf{q} = \mathbf{q}_R - \mathbf{q}_L = \alpha_1\mathbf{r}_1 + \alpha_2\mathbf{r}_2 + \cdots + \alpha_m\mathbf{r}_m. \tag{18}$$

Step 2. Each w_p has one jump, at $x = 0$, which moves with speed λ_p:

$$w_p(x,t) = \begin{cases} w_{p,L}, & x < \lambda_p t, \\ w_{p,R}, & x > \lambda_p t, \end{cases} \qquad \text{i.e.} \qquad w_p(x,t) = w_{p,L} + \begin{cases} 0, & x < \lambda_p t, \\ \alpha_p, & x > \lambda_p t. \end{cases}$$

Step 3. Recombine. Using the second form and $\sum_p w_{p,L}\mathbf{r}_p = R\mathbf{w}_L = \mathbf{q}_L$,

$$\mathbf{q}(x,t) = \sum_{p=1}^{m} w_p(x,t)\,\mathbf{r}_p = \mathbf{q}_L + \sum_{p\,:\,\lambda_p t < x} \alpha_p \mathbf{r}_p.$$

The number α_p is the **strength** of the p-th wave, and $\mathcal{W}_p = \alpha_p\mathbf{r}_p$ is the jump in $\mathbf{q}$ across it.

Riemann Solution for a Linear Hyperbolic System

The solution depends only on x/t and consists of m discontinuities travelling at speeds $\lambda_1 < \cdots < \lambda_m$, separating $m+1$ constant states:

$$\mathbf{q}(x,t) = \mathbf{q}_L + \sum_{p\,:\,\lambda_p < x/t} \mathcal{W}_p \;=\; \mathbf{q}_R - \sum_{p\,:\,\lambda_p > x/t} \mathcal{W}_p, \qquad \mathcal{W}_p = \alpha_p\mathbf{r}_p. \tag{19}$$

Crossing the p-th wave from left to right, $\mathbf{q}$ jumps by $\mathcal{W}_p$.

The second form follows from the first because $\sum_{\text{all } p} \mathcal{W}_p = \Delta\mathbf{q} = \mathbf{q}_R - \mathbf{q}_L$ by (18): the waves with $\lambda_p < x/t$ plus the waves with $\lambda_p > x/t$ add up to the full jump.

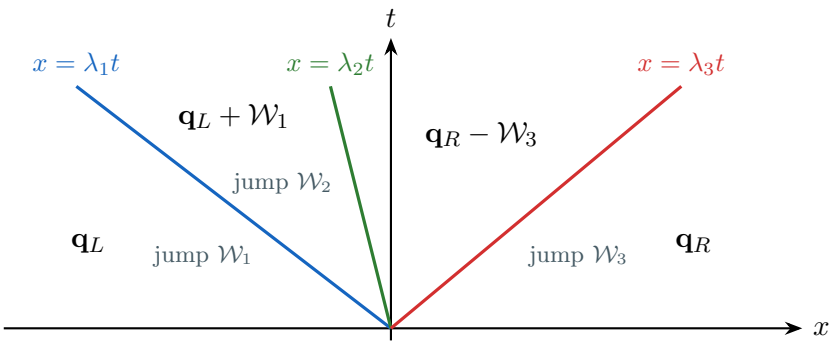


Figure 1: Structure of the Riemann solution for a linear system with $m = 3$ and $\lambda_1 < \lambda_2 < 0 < \lambda_3$. Each wave is a discontinuity carrying the jump $\mathcal{W}_p = \alpha_p \mathbf{r}_p$. The state along the t-axis ($x = 0$) is $\mathbf{q}_R - \mathcal{W}_3 = \mathbf{q}_L + \mathcal{W}_1 + \mathcal{W}_2$.

Consistency with Rankine-Hugoniot. Each wave is a discontinuity of speed $s = \lambda_p$ with jump $[\mathbf{q}] = \mathcal{W}_p$. The Rankine-Hugoniot condition of Lecture 5, $s\,[\mathbf{q}] = [\mathbf{f}(\mathbf{q})]$, with $\mathbf{f}(\mathbf{q}) = A\mathbf{q}$ gives $[\mathbf{f}] = A[\mathbf{q}]$, so it reads

$$\lambda_p \mathcal{W}_p = A\mathcal{W}_p,$$

which is exactly the statement that $\mathcal{W}_p$ is an eigenvector with eigenvalue λ_p. So for a linear system, *the only jumps that can propagate are multiples of eigenvectors, and they propagate at the eigenvalue speeds.* For nonlinear systems this picture survives, but some waves become rarefactions (Lecture 15).

5.2 The Acoustic Riemann Problem

For acoustics, $m = 2$, so there are two waves and a single intermediate state $\mathbf{q}_m = (p_m, u_m)^T$ between them. Write $\Delta p = p_R - p_L$ and $\Delta u = u_R - u_L$.

Wave strengths. From (17) and (12),

$$\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix} = R^{-1}\begin{pmatrix}\Delta p\\ \Delta u\end{pmatrix} = \frac{1}{2Z_0}\begin{pmatrix}-1 & Z_0\\ 1 & Z_0\end{pmatrix}\begin{pmatrix}\Delta p\\ \Delta u\end{pmatrix} = \frac{1}{2Z_0}\begin{pmatrix}-\Delta p + Z_0\Delta u\\ \Delta p + Z_0\Delta u\end{pmatrix}.$$

Middle state. Crossing the first wave from the left, $\mathbf{q}_m = \mathbf{q}_L + \alpha_1 \mathbf{r}_1$. Componentwise:

$$p_m = p_L + \alpha_1(-Z_0) = p_L - \frac{-\Delta p + Z_0\Delta u}{2} = p_L + \frac{p_R - p_L}{2} - \frac{Z_0}{2}\Delta u = \frac{p_L + p_R}{2} - \frac{Z_0}{2}(u_R - u_L),$$
$$u_m = u_L + \alpha_1(1) = u_L + \frac{-\Delta p + Z_0\Delta u}{2Z_0} = u_L + \frac{u_R - u_L}{2} - \frac{\Delta p}{2Z_0} = \frac{u_L + u_R}{2} - \frac{1}{2Z_0}(p_R - p_L).$$

As a check, we can also come from the right, $\mathbf{q}_m = \mathbf{q}_R - \alpha_2 \mathbf{r}_2$:

$$p_m = p_R - Z_0\alpha_2 = p_R - \frac{\Delta p + Z_0\Delta u}{2} = \frac{p_L + p_R}{2} - \frac{Z_0}{2}\Delta u,$$
$$u_m = u_R - \alpha_2 = u_R - \frac{\Delta p + Z_0\Delta u}{2Z_0} = \frac{u_L + u_R}{2} - \frac{\Delta p}{2Z_0}. \checkmark$$

Middle State of the Acoustic Riemann Problem

$$p_m = \frac{p_L + p_R}{2} - \frac{Z_0}{2}(u_R - u_L), \qquad u_m = \frac{u_L + u_R}{2} - \frac{1}{2Z_0}(p_R - p_L). \tag{20}$$

The waves travel at $\lambda_1 = u_0 - c_0$ and $\lambda_2 = u_0 + c_0$; the middle state does not depend on u_0.

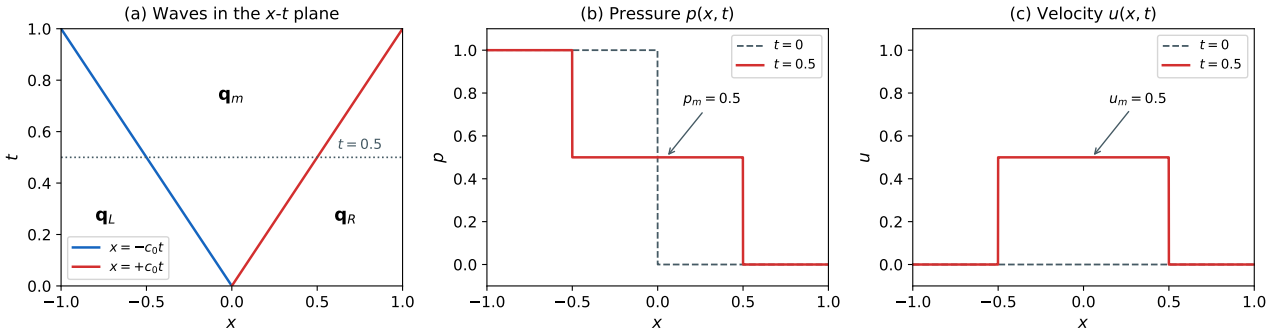


Figure 2: The acoustic Riemann problem with $\mathbf{q}_L = (1, 0)$, $\mathbf{q}_R = (0, 0)$, $\rho_0 = c_0 = 1$, $u_0 = 0$. (a) The two waves $x = \pm c_0 t$ separate the states $\mathbf{q}_L$, $\mathbf{q}_m$, $\mathbf{q}_R$. (b), (c) Pressure and velocity at $t = 0.5$.

A second derivation with Riemann invariants. Without any matrices: $p + Z_0 u$ travels to the right with the 2-wave, so the middle state inherits it from the *left* state; $p - Z_0 u$ travels to the left with the 1-wave, so the middle state inherits it from the *right* state:

$$p_m + Z_0 u_m = p_L + Z_0 u_L, \qquad p_m - Z_0 u_m = p_R - Z_0 u_R.$$

Adding the two equations and dividing by 2 gives p_m; subtracting them and dividing by $2Z_0$ gives u_m. The result is again (20).

5.3 Worked Example 1: A Pressure Jump (Linear Shock Tube)

Take $\rho_0 = 1$, $c_0 = 1$, so $Z_0 = \rho_0 c_0 = 1$ and $K_0 = 1$, background at rest ($u_0 = 0$), and

$$\mathbf{q}_L = (p_L, u_L) = (1, 0), \qquad \mathbf{q}_R = (p_R, u_R) = (0, 0).$$

Speeds: $\lambda_1 = 0 - 1 = -1$, $\lambda_2 = 0 + 1 = 1$.

Jumps: $\Delta p = 0 - 1 = -1$, $\Delta u = 0 - 0 = 0$.

Strengths: $\alpha_1 = \frac{-(-1)+1\cdot 0}{2\cdot 1} = \frac{1}{2}$, $\alpha_2 = \frac{-1+1\cdot 0}{2\cdot 1} = -\frac{1}{2}$.

Check: $\alpha_1 \mathbf{r}_1 + \alpha_2 \mathbf{r}_2 = \frac{1}{2}(-1, 1)^T - \frac{1}{2}(1, 1)^T = (-1, 0)^T = \Delta\mathbf{q}$. ✓

Middle state:

$$\mathbf{q}_m = \mathbf{q}_L + \alpha_1 \mathbf{r}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

Or directly from (20): $p_m = \frac{1+0}{2} - \frac{1}{2}(0 - 0) = \frac{1}{2}$ and $u_m = \frac{0+0}{2} - \frac{1}{2}(0 - 1) = \frac{1}{2}$.

Solution:

$$\mathbf{q}(x, t) = \begin{cases} (1, 0), & x < -t, \\ (\frac{1}{2}, \frac{1}{2}), & -t < x < t, \\ (0, 0), & x > t. \end{cases}$$

The high pressure on the left pushes the fluid to the right: in the middle region the pressure is the average and the fluid moves with $u_m = \frac{1}{2}$. The solution is shown in Figure 2. This is the linear analogue of the *shock tube* problem that we will solve for the full Euler equations in Module 6.

5.4 Worked Example 2: Colliding Streams

Same fluid ($Z_0 = 1$, $u_0 = 0$), but now two streams at equal pressure rush towards each other:

$$\mathbf{q}_L = (0, 1), \qquad \mathbf{q}_R = (0, -1).$$

Jumps: $\Delta p = 0$, $\Delta u = -1 - 1 = -2$.

Strengths: $\alpha_1 = \frac{-0+1\cdot(-2)}{2} = -1$, $\alpha_2 = \frac{0+1\cdot(-2)}{2} = -1$.

Middle state: $\mathbf{q}_m = \mathbf{q}_L + \alpha_1 \mathbf{r}_1 = (0, 1)^T + (-1)(-1, 1)^T = (1, 0)^T$. Check with (20): $p_m = 0 - \frac{1}{2}(-1 - 1) = 1$, $u_m = 0 - \frac{1}{2}(0) = 0$. ✓

The fluid in the middle is brought to rest ($u_m = 0$), and its kinetic energy is converted into a pressure rise $p_m = 1 = Z_0\,|u_L|$. Two compression waves move outwards at speed c_0. Notice the formula $p_m = Z_0|u_L|$: the impedance Z_0 measures how much pressure is needed to stop a stream. This is the linear version of the pressure rise when a stream hits a wall (water hammer).

6 Godunov's Method for Linear Systems

6.1 The Godunov Flux

Godunov's method (Lecture 10) for $\mathbf{q}_t + \mathbf{f}(\mathbf{q})_x = 0$ is

$$\mathbf{q}_i^{n+1} = \mathbf{q}_i^n - \frac{\Delta t}{\Delta x}\big(\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2}\big), \qquad \mathbf{F}_{i+1/2} = \mathbf{f}\big(\mathbf{q}^*(\mathbf{q}_i^n, \mathbf{q}_{i+1}^n)\big), \tag{21}$$

where $\mathbf{q}^*(\mathbf{q}_L, \mathbf{q}_R)$ is the value of the Riemann solution on the interface itself, i.e. at $x/t = 0$. We now compute this flux for $\mathbf{f}(\mathbf{q}) = A\mathbf{q}$, in four short steps.

Step 1: the interface state. Put $x/t = 0$ in (19). The waves to the left of the interface are those with $\lambda_p < 0$:

$$\mathbf{q}^* = \mathbf{q}_L + \sum_{p\,:\,\lambda_p<0} \mathcal{W}_p.$$

Step 2: the flux. Multiply by A and use $A\mathcal{W}_p = \lambda_p \mathcal{W}_p$ (each $\mathcal{W}_p$ is an eigenvector):

$$\mathbf{F} = A\mathbf{q}^* = A\mathbf{q}_L + \sum_{p\,:\,\lambda_p<0} \lambda_p \mathcal{W}_p.$$

Step 3: write the sum as a matrix. Define the negative and positive parts of each speed,

$$\lambda_p^- = \min(\lambda_p, 0), \qquad \lambda_p^+ = \max(\lambda_p, 0), \qquad \lambda_p = \lambda_p^+ + \lambda_p^-, \qquad |\lambda_p| = \lambda_p^+ - \lambda_p^-.$$

Since $\lambda_p^- = 0$ for the waves with $\lambda_p \geq 0$, we may sum over all p:

$$\sum_{p\,:\,\lambda_p<0} \lambda_p \mathcal{W}_p = \sum_{p=1}^{m} \lambda_p^- \alpha_p \mathbf{r}_p = R\,\Lambda^-\boldsymbol{\alpha} = R\,\Lambda^- R^{-1}\,\Delta\mathbf{q} = A^-\Delta\mathbf{q},$$

where we used $\boldsymbol{\alpha} = R^{-1}\Delta\mathbf{q}$ and defined the matrices

$$A^\pm = R\,\Lambda^\pm R^{-1}, \qquad \Lambda^\pm = \mathrm{diag}(\lambda_1^\pm, \ldots, \lambda_m^\pm), \qquad |A| = R\,|\Lambda|\,R^{-1}. \tag{22}$$

A^+ keeps only the right-going waves and A^- only the left-going ones. Because $\Lambda = \Lambda^+ + \Lambda^-$ and $|\Lambda| = \Lambda^+ - \Lambda^-$,

$$A = A^+ + A^-, \qquad |A| = A^+ - A^-.$$

So $\mathbf{F} = A\mathbf{q}_L + A^-\Delta\mathbf{q}$.

Step 4: other forms. Starting from the right instead, $\mathbf{q}^* = \mathbf{q}_R - \sum_{\lambda_p>0} \mathcal{W}_p$, and the same steps give $\mathbf{F} = A\mathbf{q}_R - A^+\Delta\mathbf{q}$. Then:

- From the left form: $\mathbf{F} = A\mathbf{q}_L + A^-(\mathbf{q}_R - \mathbf{q}_L) = (A - A^-)\mathbf{q}_L + A^-\mathbf{q}_R = A^+\mathbf{q}_L + A^-\mathbf{q}_R$.
- Averaging the left and right forms: $\mathbf{F} = \frac{1}{2}(A\mathbf{q}_L + A\mathbf{q}_R) + \frac{1}{2}(A^- - A^+)\Delta\mathbf{q} = \frac{1}{2}A(\mathbf{q}_L + \mathbf{q}_R) - \frac{1}{2}|A|\,\Delta\mathbf{q}$.

Godunov Flux for $\mathbf{q}_t + A\mathbf{q}_x = 0$

$$\begin{aligned}\mathbf{F}(\mathbf{q}_L, \mathbf{q}_R) = A^+\mathbf{q}_L + A^-\mathbf{q}_R \;&=\; A\mathbf{q}_L + A^-\Delta\mathbf{q} \;=\; A\mathbf{q}_R - A^+\Delta\mathbf{q} \\ &= \tfrac{1}{2}A(\mathbf{q}_L + \mathbf{q}_R) - \tfrac{1}{2}|A|\,(\mathbf{q}_R - \mathbf{q}_L).\end{aligned} \tag{23}$$

The first form is **flux vector splitting**: the right-going part of the flux comes from the left cell and the left-going part from the right cell. The last form is a central flux plus a matrix-valued numerical viscosity $\frac{1}{2}|A|$. For $m = 1$, $A = a$ and $|A| = |a|$, and it reduces to the scalar upwind flux $\frac{1}{2}a(u_L + u_R) - \frac{1}{2}|a|(u_R - u_L)$ of Lecture 10.

6.2 Example: The Splitting Matrices for Acoustics at Rest

Take $u_0 = 0$, so $\lambda_1 = -c_0$, $\lambda_2 = c_0$, and $\Lambda^+ = \mathrm{diag}(0, c_0)$, $\Lambda^- = \mathrm{diag}(-c_0, 0)$.

Computing A^+. First $R\Lambda^+$ (multiplying column p of R by λ_p^+), then multiply by R^{-1}:

$$R\Lambda^+ = \begin{pmatrix} -Z_0 & Z_0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & c_0 \end{pmatrix} = \begin{pmatrix} 0 & Z_0c_0 \\ 0 & c_0 \end{pmatrix},$$

$$A^+ = R\Lambda^+R^{-1} = \begin{pmatrix} 0 & Z_0c_0 \\ 0 & c_0 \end{pmatrix}\cdot\frac{1}{2Z_0}\begin{pmatrix} -1 & Z_0 \\ 1 & Z_0 \end{pmatrix} = \frac{1}{2Z_0}\begin{pmatrix} Z_0c_0 & Z_0^2c_0 \\ c_0 & Z_0c_0 \end{pmatrix} = \frac{c_0}{2}\begin{pmatrix} 1 & Z_0 \\ 1/Z_0 & 1 \end{pmatrix}.$$

Computing A^-. Using (8), $A = \begin{pmatrix} 0 & Z_0c_0 \\ c_0/Z_0 & 0 \end{pmatrix} = c_0\begin{pmatrix} 0 & Z_0 \\ 1/Z_0 & 0 \end{pmatrix}$, so

$$A^- = A - A^+ = c_0\begin{pmatrix} 0 & Z_0 \\ 1/Z_0 & 0 \end{pmatrix} - \frac{c_0}{2}\begin{pmatrix} 1 & Z_0 \\ 1/Z_0 & 1 \end{pmatrix} = \frac{c_0}{2}\begin{pmatrix} -1 & Z_0 \\ 1/Z_0 & -1 \end{pmatrix}.$$

Computing $|A|$.

$$|A| = A^+ - A^- = \frac{c_0}{2}\begin{pmatrix} 1-(-1) & Z_0 - Z_0 \\ 1/Z_0 - 1/Z_0 & 1-(-1) \end{pmatrix} = c_0\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = c_0\, I.$$

This is no accident: both speeds have the same magnitude $|\lambda_p| = c_0$, so $|\Lambda| = c_0 I$ and $|A| = R(c_0 I)R^{-1} = c_0 I$.

The flux for Example 1. With $\mathbf{q}_L = (1, 0)$, $\mathbf{q}_R = (0, 0)$, $Z_0 = c_0 = 1$:

$$\mathbf{F} = A^+\mathbf{q}_L + A^-\mathbf{q}_R = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \mathbf{0} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

Compare with $A\mathbf{q}^*$ where $\mathbf{q}^* = \mathbf{q}_m = (\frac{1}{2}, \frac{1}{2})$ (the interface lies in the middle region because $\lambda_1 < 0 < \lambda_2$): $A\mathbf{q}_m = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$. ✓

6.3 Wave-Propagation Form

For cell i, write $\Delta\mathbf{q}_{i-1/2} = \mathbf{q}_i - \mathbf{q}_{i-1}$ and $\Delta\mathbf{q}_{i+1/2} = \mathbf{q}_{i+1} - \mathbf{q}_i$ (time level n). At the right interface $x_{i+1/2}$, cell i is the *left* state, so we use the form $\mathbf{F} = A\mathbf{q}_L + A^-\Delta\mathbf{q}$:

$$\mathbf{F}_{i+1/2} = A\mathbf{q}_i + A^-\Delta\mathbf{q}_{i+1/2}.$$

At the left interface $x_{i-1/2}$, cell i is the *right* state, so we use $\mathbf{F} = A\mathbf{q}_R - A^+\Delta\mathbf{q}$:

$$\mathbf{F}_{i-1/2} = A\mathbf{q}_i - A^+\Delta\mathbf{q}_{i-1/2}.$$

Subtracting, the term $A\mathbf{q}_i$ cancels: $\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2} = A^+\Delta\mathbf{q}_{i-1/2} + A^-\Delta\mathbf{q}_{i+1/2}$. Hence

$$\mathbf{q}_i^{n+1} = \mathbf{q}_i^n - \frac{\Delta t}{\Delta x}\Big(A^+\Delta\mathbf{q}_{i-1/2} + A^-\Delta\mathbf{q}_{i+1/2}\Big). \tag{24}$$

This is LeVeque's **wave-propagation form**: cell i is updated only by the right-going waves from its left interface and the left-going waves from its right interface.

6.4 Each Wave is Upwinded Separately

Multiply (24) from the left by R^{-1}, and set $\mathbf{w}_i = R^{-1}\mathbf{q}_i$. Since $R^{-1}A^{\pm} = R^{-1}R\Lambda^{\pm}R^{-1} = \Lambda^{\pm}R^{-1}$,

$$R^{-1}A^{\pm}\Delta\mathbf{q}_{i\mp1/2} = \Lambda^{\pm}\,R^{-1}\Delta\mathbf{q}_{i\mp1/2} = \Lambda^{\pm}\,\Delta\mathbf{w}_{i\mp1/2}.$$

$\Lambda^{\pm}$ is diagonal, so row p reads, with $\nu_p^{\pm} = \lambda_p^{\pm}\Delta t/\Delta x$,

$$w_{p,i}^{n+1} = w_{p,i}^{n} - \nu_p^{+}\left(w_{p,i}^{n} - w_{p,i-1}^{n}\right) - \nu_p^{-}\left(w_{p,i+1}^{n} - w_{p,i}^{n}\right). \tag{25}$$

If $\lambda_p > 0$, only the first difference survives (backward difference); if $\lambda_p < 0$, only the second (forward difference). This is exactly the first-order upwind scheme of Lecture 6 for $(w_p)_t + \lambda_p(w_p)_x = 0$.

Insight: Godunov for a Linear System

Godunov's method for $\mathbf{q}_t + A\mathbf{q}_x = 0$ is equivalent to: transform to characteristic variables, apply the scalar upwind scheme to each w_p with its own speed λ_p, transform back. We never have to perform the transformation in the code: the matrices $A^{\pm}$ do it for us.

6.5 Stability: the CFL Condition

By (25) it suffices to study one scalar upwind scheme. Take $\lambda_p > 0$ and write $\nu = \nu_p^{+}$: $w_i^{n+1} = (1-\nu)\,w_i^n + \nu\,w_{i-1}^n$. Von Neumann analysis (Lecture 7) with $w_i^n = G^n e^{\mathrm{i}i\theta}$:

$$G = 1 - \nu + \nu e^{-\mathrm{i}\theta}, \qquad |G|^2 = (1-\nu+\nu\cos\theta)^2 + \nu^2\sin^2\theta = 1 - 2\nu(1-\nu)(1-\cos\theta).$$

Since $1-\cos\theta \geq 0$, we have $|G| \leq 1$ for all θ exactly when $\nu(1-\nu) \geq 0$, i.e. $0 \leq \nu \leq 1$. The case $\lambda_p < 0$ is the mirror image. Requiring this for every wave:

$$\text{CFL condition:} \qquad \max_p |\lambda_p|\,\frac{\Delta t}{\Delta x} \leq 1. \tag{26}$$

The time step is limited by the *fastest* wave. For acoustics this is $|u_0| + c_0$, and for the Euler equations $\max(|u| + c)$.

6.6 The Rusanov Flux for Systems

The Rusanov (local Lax-Friedrichs) flux of Lecture 10 extends to systems by replacing $|f'(u)|$ with the largest wave speed $s_{\max} = \max_p |\lambda_p|$ (the spectral radius of A):

$$\mathbf{F}^{\mathrm{Rus}}(\mathbf{q}_L, \mathbf{q}_R) = \tfrac{1}{2}\big(\mathbf{f}(\mathbf{q}_L) + \mathbf{f}(\mathbf{q}_R)\big) - \tfrac{1}{2}s_{\max}\,(\mathbf{q}_R - \mathbf{q}_L). \tag{27}$$

It needs no eigenvectors, only an upper bound on the wave speeds, and this is why it is so popular for complicated systems. To compare with Godunov, subtract the last form of (23); the central parts cancel:

$$\mathbf{F}^{\mathrm{Rus}} - \mathbf{F}^{\mathrm{God}} = -\tfrac{1}{2}s_{\max}\Delta\mathbf{q} + \tfrac{1}{2}|A|\Delta\mathbf{q} = -\tfrac{1}{2}\big(s_{\max}I - |A|\big)\Delta\mathbf{q}.$$

Writing $I = RR^{-1}$ and $|A| = R|\Lambda|R^{-1}$, the matrix in brackets is

$$s_{\max}I - |A| = R\ \mathrm{diag}\left(s_{\max} - |\lambda_1|, \ldots, s_{\max} - |\lambda_m|\right) R^{-1}.$$

So the Rusanov flux gives *every* wave the numerical viscosity of the fastest one. The extra viscosity on wave p is proportional to $s_{\max} - |\lambda_p| \geq 0$: zero for the fastest wave, largest for the slowest.

Insight: When Does the Extra Diffusion Matter?

For acoustics at rest ($u_0 = 0$) we computed $|A| = c_0 I$ and $s_{\max} = c_0$, so $s_{\max}I - |A| = 0$: the Rusanov and Godunov fluxes are *identical*. With a background flow they differ. For example, with $c_0 = 1$, $u_0 = 0.5$: $|\lambda_1| = 0.5$, $|\lambda_2| = 1.5 = s_{\max}$, and the slow wave gets extra viscosity proportional to $1.5 - 0.5 = 1$. For the Euler equations the worst case is the contact wave, with speed u, next to sound waves with speed $|u| + c$: at low Mach number Rusanov smears contacts badly. This is the motivation for the Roe, HLL and HLLC solvers of Module 6, which resolve the individual waves.

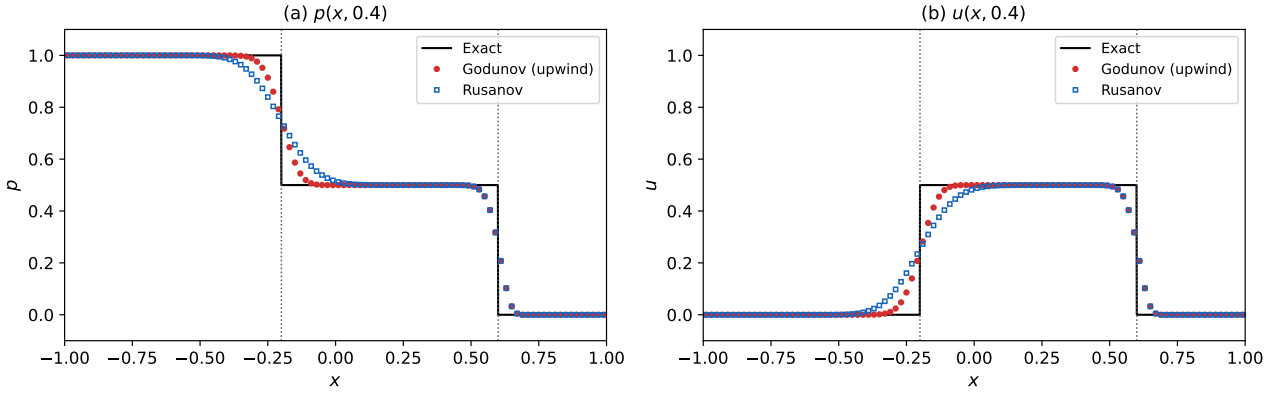


Figure 3: Acoustics with $u_0 = 0.5$, $\rho_0 = c_0 = 1$, pressure-jump Riemann data, $N = 100$, CFL 0.9, $t = 0.4$. Dotted lines: exact wave positions $x = \lambda_1 t = -0.2$ and $x = \lambda_2 t = 0.6$.

7 Numerical Results

We solve the pressure-jump Riemann problem of Example 1 on $[-1, 1]$, now with a background flow $u_0 = 0.5$ ($\rho_0 = c_0 = 1$). Before running, let us predict everything we can by hand:

- *Speeds:* $\lambda_1 = 0.5 - 1 = -0.5$, $\lambda_2 = 0.5 + 1 = 1.5$, so $s_{\max} = 1.5$.
- *Middle state:* unchanged, $\mathbf{q}_m = (\frac{1}{2}, \frac{1}{2})$, since (20) does not involve u_0.
- *Wave positions at $t = 0.4$:* $\lambda_1 t = -0.2$ and $\lambda_2 t = 0.6$.
- *Time step:* with CFL number 0.9, $\Delta t = 0.9\,\Delta x / s_{\max} = 0.6\,\Delta x$. The local Courant numbers are then $|\lambda_1|\Delta t/\Delta x = 0.5 \times 0.6 = 0.3$ for the slow wave and $1.5 \times 0.6 = 0.9$ for the fast wave.

Figure 3 shows the result with $N = 100$ cells.

At the fast wave ($x = 0.6$, local Courant number 0.9) the two schemes coincide and the jump is sharp. At the slow wave ($x = -0.2$, local Courant number 0.3) Godunov smears the jump over a few cells, and Rusanov over noticeably more. (Recall from Lecture 8 that the numerical diffusion of the upwind scheme is $\frac{1}{2}|\lambda|\Delta x(1 - \nu)$, largest for small Courant numbers.) The L^1 errors in p are:

N	100	200	400	800
Godunov	3.48×10^{-2}	2.47×10^{-2}	1.75×10^{-2}	1.23×10^{-2}
Rusanov	5.53×10^{-2}	3.91×10^{-2}	2.76×10^{-2}	1.95×10^{-2}

Each time N doubles, both errors are divided by about $1.41 = \sqrt{2}$; for instance $3.48/2.47 \approx 1.41$. So the error behaves like $\Delta x^{1/2}$, the usual rate of a first-order scheme at a discontinuity. The ratio Rusanov/Godunov is $5.53/3.48 \approx 1.59$ on every grid: Rusanov is about 60% less accurate. With $u_0 = 0$ the two schemes agree to machine precision, as predicted.

8 Looking Ahead

For a nonlinear system the Jacobian $A(\mathbf{q})$ depends on the solution, and three things change:

1. Waves can be **shocks** or **rarefactions**, as for Burgers' equation, or **contact discontinuities**, as for linear advection. The jumps no longer lie exactly along eigenvectors (Lecture 15).
2. The exact Riemann solution requires solving nonlinear algebraic equations. **Approximate Riemann solvers** replace it: Roe's solver picks a matrix $\hat{A}(\mathbf{q}_L, \mathbf{q}_R)$ and uses the linear Godunov flux (23) with it.
3. For high-order schemes (MUSCL, WENO), reconstruction should be done in the **characteristic variables** $\mathbf{w} = R^{-1}\mathbf{q}$, computed locally, so that limiters act on one wave at a time.

9 Summary

Concept	Statement	Significance
Flux Jacobian	$A(\mathbf{q}) = \mathbf{f}'(\mathbf{q})$, quasi-linear form $\mathbf{q}_t + A\mathbf{q}_x = 0$	Replaces $f'(u)$ of the scalar case
Hyperbolicity	$A = R\Lambda R^{-1}$ with real λ_p, complete eigenvectors	Waves with real, finite speeds
Characteristic variables	$\mathbf{w} = R^{-1}\mathbf{q}$, $(w_p)_t + \lambda_p (w_p)_x = 0$	m decoupled advection equations
Acoustics	$\lambda = u_0 \mp c_0$, $\mathbf{r} = (\mp Z_0, 1)^T$	Model for the sound waves of Euler
Riemann solution	$\Delta\mathbf{q} = \sum \alpha_p \mathbf{r}_p$; jump $\mathcal{W}_p = \alpha_p \mathbf{r}_p$ at speed λ_p	Building block of Godunov's method
Godunov flux	$A^+\mathbf{q}_L + A^-\mathbf{q}_R = \frac{1}{2}A(\mathbf{q}_L + \mathbf{q}_R) - \frac{1}{2}\lvert A\rvert \Delta\mathbf{q}$	Upwind in every wave family
CFL	$\max_p \lvert\lambda_p\rvert \, \Delta t/\Delta x \le 1$	Fastest wave limits Δt
Rusanov flux	Viscosity $\frac{1}{2}s_{\max}$ for all waves	Simple, but smears slow waves

10 Exercises

Exercises for Lecture 14

E1. Hyperbolic or Not?

(a) Decide whether $\mathbf{q}_t + A\mathbf{q}_x = 0$ is strictly hyperbolic, hyperbolic or not hyperbolic for $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

(b) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, find $\lambda_1 < \lambda_2$, R, R^{-1} and the characteristic variables. Solve the Riemann problem with $\mathbf{q}_L = (1, 0)$, $\mathbf{q}_R = (0, 0)$ and sketch both components at $t = 1$.

E2. Acoustics:

(a) Verify $A\mathbf{r}_2 = \lambda_2 \mathbf{r}_2$ for (12) by multiplying out, and verify $R^{-1}R = I$.

(b) Show directly from (7) that $(p + Z_0 u)_t + (u_0 + c_0)(p + Z_0 u)_x = 0$.

(c) Show that the acoustic system with $u_0 = 0$ also implies $u_{tt} = c_0^2 u_{xx}$.

(d) Show that $E = \frac{1}{2K_0}p^2 + \frac{\rho_0}{2}u^2$ satisfies $E_t + \big(u_0 E + p\,u\big)_x = 0$. What does this say about $\int E\,dx$ on a periodic domain?

E3. Riemann Problems by Hand: Take $\rho_0 = 1$, $c_0 = 2$, $u_0 = 0$.

(a) $\mathbf{q}_L = (2, 1)$, $\mathbf{q}_R = (1, 0)$. Find Z_0, α_1, α_2 and $\mathbf{q}_m$, and check that $\alpha_1 \mathbf{r}_1 + \alpha_2 \mathbf{r}_2 = \Delta\mathbf{q}$. Sketch p and u at $t = 0.25$.

(b) *Separating streams:* $\mathbf{q}_L = (0, -1)$, $\mathbf{q}_R = (0, 1)$. Find $\mathbf{q}_m$ and explain the sign of p_m physically.

(c) Repeat (a) with $u_0 = 3$. Which quantities change, and which do not? Where are the waves at $t = 0.25$?

E4. Forms of the Godunov Flux:

(a) Starting from $\mathbf{q}^* = \mathbf{q}_R - \sum_{\lambda_p > 0} \mathcal{W}_p$, show that $\mathbf{F} = A\mathbf{q}_R - A^+\Delta\mathbf{q}$.

(b) For acoustics with $u_0 > c_0$ (supersonic), show that $A^+ = A$ and $A^- = 0$, so that $\mathbf{F} = A\mathbf{q}_L$. Interpret.

(c) For acoustics with $u_0 = 0$, use the matrices of Section 6.2 to show that the first component of $\mathbf{F}(\mathbf{q}_L, \mathbf{q}_R)$ equals $K_0 u_m$ and the second equals p_m/ρ_0, with (p_m, u_m) from (20).

(d) Show that the extra viscosity matrix $s_{\max} I - |A|$ of the Rusanov flux has non-negative eigenvalues.

E5. Shallow Water (preview of Lecture 15): The shallow water equations are $h_t + (hu)_x = 0$, $(hu)_t + \left(hu^2 + \frac{1}{2}gh^2\right)_x = 0$.

(a) With $\mathbf{q} = (q_1, q_2) = (h, hu)$, write the flux as a function of q_1, q_2 (hint: $hu^2 = q_2^2/q_1$), compute the Jacobian $A(\mathbf{q})$, and show that its eigenvalues are $u \pm \sqrt{gh}$. When is the system strictly hyperbolic?

(b) Linearize about (h_0, u_0) and compare with acoustics. What plays the role of c_0?

E6. Euler Eigenvalues: In the primitive variables $\mathbf{V} = (\rho, u, p)^T$ the one-dimensional Euler equations read $\mathbf{V}_t + B(\mathbf{V})\mathbf{V}_x = 0$ with $B = \begin{pmatrix} u & \rho & 0 \\ 0 & u & 1/\rho \\ 0 & \rho c^2 & u \end{pmatrix}$. Expand $\det(B - \lambda I)$ along the first column to show that the eigenvalues are $u - c$, u, $u + c$, and find the eigenvectors. Which variables jump across the middle wave?

E7. Computational Exercise: Run `acoustics_godunov.jl` from the course website.

(a) Set $u_0 = 0$ and confirm that the Godunov and Rusanov solutions coincide.

(b) Set $u_0 = 1$ (sonic: $\lambda_1 = 0$) and $u_0 = 1.5$ (supersonic). Describe the solutions. In the supersonic case, check numerically that the Godunov flux is simply $A\mathbf{q}_L$.